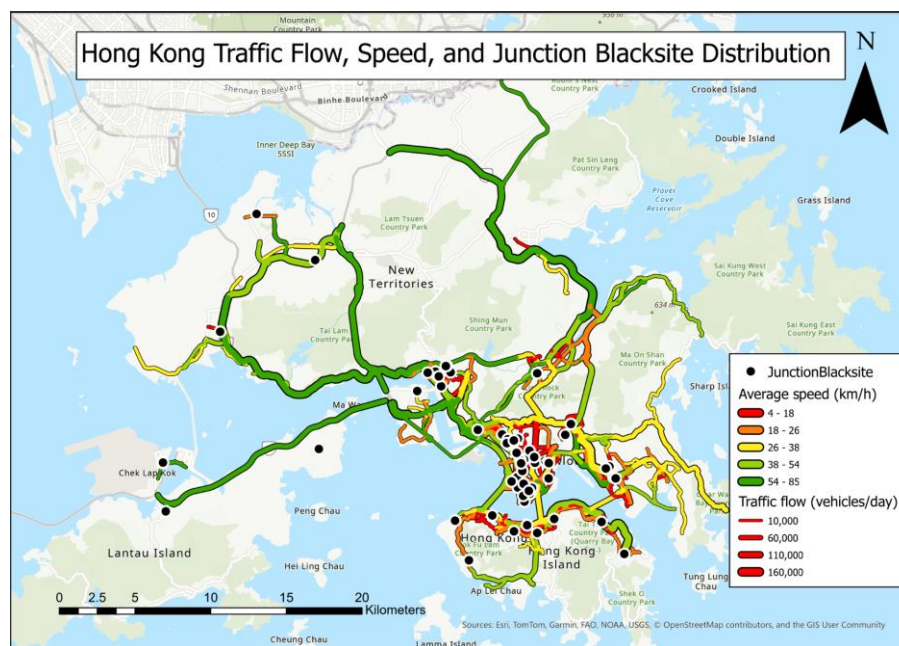


LSGI3260A – Cartography and Geo-visualization

Group ■ - Group Report

Subject Lecturer: Dr. Sissi Chen

Title: Hong Kong Traffic Flow, Speed, and Junction Black site Distribution



Students

Category	Percentage	Category	Percentage
Very important	35%	Very important	35%
Important	45%	Important	45%
Not important	20%	Not important	20%
Don't know	10%	Don't know	10%
Very important	35%	Very important	35%
Important	45%	Important	45%
Not important	20%	Not important	20%
Don't know	10%	Don't know	10%

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Introduction

This project applies cartographic principles and produce a thematic map of Hong Kong's road network, including traffic flow volumes, average speeds, junction blacksite locations. This project processes and analyzes data to reveal meaningful spatial patterns, and evaluate the effectiveness of visual communication.

Project Rationale

Hong Kong faces frequent traffic congestion and road safety issues due to high population density, high road usage, and limited road space. According to TomTom Traffic Index (2025), the average congestion level is 47.7%, and the traffic speed for peak hours is less than 25 km/h. According to the Transport Department (TD, 2024), the average daily vehicle-kilometer is 39.69 million. This shows that Hong Kong has high traffic flow in general, with some areas with traffic slowdown, and the high relevance of this topic to daily lives.

According to TD (2021), junction blacksites are locations with 6+ pedestrian injury accidents or 9+ casualty accidents annually. We believe that junction blacksites have some correlation to the location of the roads, speeds, and traffic flow. These insights may support urban planning, traffic management, and safety interventions.

Project Objectives and Deliverables

The first objective of this group project is to apply the cartographic principles in the thematic map production, achieving effective visual communication with the target audience. We also aim to provide insights for urban planning and traffic management. By showing the correlations between traffic flow, average speed, and junction blacksite locations, the map might assist decision making and provide suggestions to future planning, ultimately improve the traffic safety and efficiency.

Intended Audience

There are multiple governmental departments that are responsible for urban development and planning, especially for the road transportation, such as the Transport Department and the Highway Department that may use maps to analyze the traffic flow and speed situation, and aim to improve the road efficiency and safety. Our map may assist them for decision making and provide insights and measures to address traffic congestion and safety issues. There are also other map users, including construction consultants and contractors, commuters, our map might provide information on the roads that are more likely to have slowdowns and accidents, and thus increase the awareness of road use in the affected regions.

Map Theme and Thematic Map Types

Map Title: Hong Kong Traffic Flow, Speed, and Junction Blacksite Distribution

The thematic map theme of our project is a flow map. Since the roads are bidirectional, no arrows were used. The line width reflects the traffic flow, the hue (transition from red to green) reflects the traffic speed. The secondary thematic map theme is a dot map. The locations of the junction blacksites are sparse and discrete data, thus a dot map is feasible for representing the junction blacksites.

Geographic coverage

Our map covers the entirety of Hong Kong, showing the highways and main roads connecting New Territories, Kowloon, and Hong Kong Island. Some roads were unavailable due to either lacking traffic flow, speed, or unavailable in the shape file (e.g. San Tin Highway)

Data sources and Methodology

1. Data Sources

For this project, multiple official data sources were used, such as the Open Data Portal (data.gov.hk) developed by Government Logistics Department (GLD), and the CSDI Portal developed by the Lands Department. Data were sourced exclusively from official open portals to ensure reliability, and also available in Appendix A. The following are the datasets used:

- Annual Traffic Census traffic data (AADT, vehicles per day) from data.gov.hk:
https://data.gov.hk/en-data/dataset/hk-td-tis_7-traffic-flow-census/resource/c3392d0f-9a7c-41ad-9b0c-cfe8c8334f51
- Road centerline shapefile (including road shapes and English names) from data.gov.hk: https://data.gov.hk/en-data/dataset/hk-td-tis_7-traffic-flow-census/resource/a2009821-77c7-4788-815f-932d595bc4fb
- Hong Kong car journey time data (travel time and road length for speed calculation) from data.gov.hk: https://data.gov.hk/en-data/dataset/hk-td-tis_34-car-journey-time-data
- Junction Blacksite list from the CSDI Portal (Transport Department):
https://portal.csd.gov.hk/geoportal/?lang=en&datasetId=td_rcd_1638947917303_26860

2. Data Processing and Cleaning for Traffic Speed Data

Average Speed Data

For average speed data, raw journey timetables were in different spreadsheets, we have merged the data from the 4 spreadsheets, summing the distance (km) and journey time (minutes) for all duplicated fields. Average speed (km/h) was calculated as $(\text{length} * 60) / \text{time}$. Then we extracted the road name and average travel speed into a new spreadsheet, converting the road names into uppercase to match the name in the shape file fields. The spreadsheet is available in the Appendix.

Road/Street	Distance (km)		Journey Time (min)		Average d (km)	Average t (h)	Average tra (km/h)
	A to B	B to A	A to B	B to A			
Aberdeen	1.60	1.70	1.90	1.90	1.65	0.031667	52.10526
Aberdeen	2.90	3.10	5.90	3.10	3	0.075	40
Airport Rd	0.70	0.00	1.00	0.00	0.7	0.016667	42
Albany Rd	0.20	0.00	3.50	0.00	0.2	0.058333	3.428571
Ap Lei Chi	1.90	1.30	3.10	1.90	1.6	0.041667	38.4
Argyle St	2.20	2.70	6.50	10.70	2.45	0.143333	17.09302
Austin Rd	1.00	1.00	5.20	3.40	1	0.071667	13.95349
Austin Rd	0.90	0.90	3.10	1.40	0.9	0.0375	24
Becher's	0.00	1.00	0.00	5.10	1	0.085	11.76471
Berwick St	0.30	0.00	2.30	0.00	0.3	0.038333	7.826087
Bonham Rd	1.40	0.60	5.30	3.20	1	0.070833	14.11765
Boundary	2.50	0.00	8.40	0.00	2.5	0.14	17.85714
Bulkeley	0.20	0.00	1.00	0.00	0.2	0.016667	12
Bute Street	0.10	0.00	0.20	0.00	0.1	0.003333	30
Butterfly	0.50	0.00	1.40	0.00	0.5	0.023333	21.42857
Cadogan	10.80	0.30	2.20	1.30	0.55	0.029167	18.85714
Caine Rd	0.90	0.00	4.30	0.00	0.9	0.071667	12.55814
Canal Rd	2.00	1.60	4.80	2.80	1.8	0.063333	28.42105
Canton Rd	0.70	1.20	2.90	4.60	0.95	0.0625	15.2
Castle Pea	0.70	2.30	0.90	8.40	1.5	0.0775	19.35484
Castle Pea	1.30	1.30	3.00	2.70	1.3	0.0475	27.36842
Castle Pea	4.80	4.90	7.40	7.20	4.85	0.121667	39.86301
Castle Pea	1.10	1.10	2.70	1.70	1.1	0.036667	30
Castle Pea	0.90	0.90	1.00	0.90	0.9	0.015833	56.84211
Castle Pea	1.90	2.00	4.90	3.80	1.95	0.0725	26.89655
Castle Pea	1.70	1.70	1.80	4.90	1.7	0.055833	30.44776
Castle Pea	2.40	2.40	5.00	5.30	2.4	0.085833	27.96117
Castle Pea	4.30	4.30	4.60	4.90	4.3	0.079167	54.31579
Castle Pea	2.20	2.20	6.70	3.00	2.2	0.080833	27.21649
Castle Pea	1.80	1.80	5.40	4.40	1.8	0.081667	22.04082
Castle Pea	5.20	5.60	11.20	12.30	5.4	0.195833	27.57447
Catchick	10.40	0.00	1.90	0.00	0.4	0.031667	12.63158

Figure 1: The spreadsheet shows the average speed calculates of the first 30 rows. The full spreadsheet is linked in the Appendix.

Some roads had slightly varying lengths/times per direction, we use the averaged values. One-way roads used single values. We have converted the road names to uppercase to match with the field values in the shape files, since joining in ArcGIS Pro is case sensitive. The “Add Join” tool was used to join the shape file containing the roads in Hong Kong, with the processed table using the “Street English Name” field, creating a new “Average Speed” attribute.

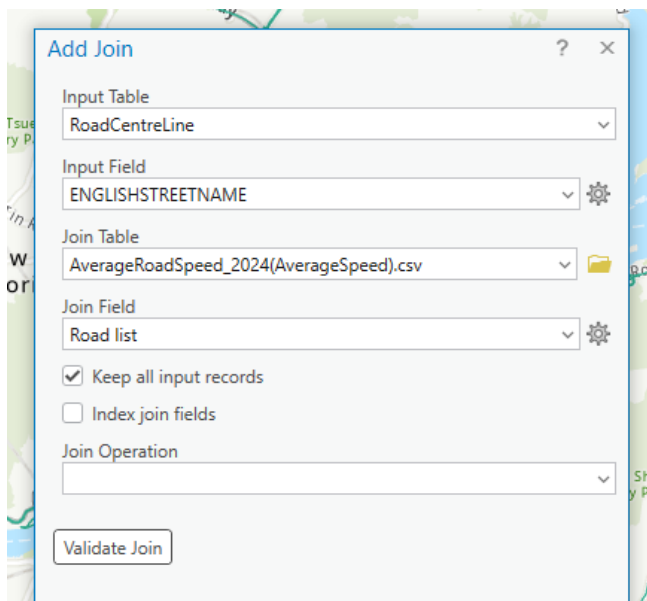


Figure 2: Using the “Add Join” functionality to add the average road speed to the map.

3. Data Processing and Cleaning for Traffic Flow Data

The raw XML (GISDB24.xml) was retrieved from the zipped data from ATC_TRAFFIC_DATA by the GLD. It shows the AADT, or annual average daily traffic, or simply vehicles per day, for each road in Hong Kong in the CurAADT tag.

```
<?xml version="1.0" encoding="UTF-8"?>
<dataroot xmlns:od="urn:schemas-microsoft-com:office:
  <PStnList>
    <ID>6605</ID>
    <Regional>NT</Regional>
    <StationNo>6605</StationNo>
    <StationType />
    <RoadType>LD</RoadType>
    <RoadNetwork>Major</RoadNetwork>
    <RoadName>Luk Keng Rd</RoadName>
    <RoadFrom>Sha Tau Kok Rd</RoadFrom>
    <RoadTo>Bride's Pool Rd</RoadTo>
    <PreAADT>830</PreAADT>
    <CurAADT>780</CurAADT>
    <ChangeAADT>-6.7</ChangeAADT>
  </PStnList>
  <PStnList>
    <ID>6606</ID>
    <Regional>NT</Regional>
    <StationNo>6606</StationNo>
    <StationType />
    <RoadType>LD</RoadType>
    <RoadNetwork>Major</RoadNetwork>
    <RoadName>Kwu Tung Rd</RoadName>
    <RoadFrom>Castle Peak Rd - Kwu Tung</RoadFrom>
    <RoadTo>Castle Peak Rd - San Tin</RoadTo>
    <PreAADT>4,480</PreAADT>
    <CurAADT>4,060</CurAADT>
    <ChangeAADT>-9.4</ChangeAADT>
  </PStnList>
</dataroot>
```

Figure 3: 2 XML entries in GISDB24.xml. Our fields of interest were only RoadName and CurAADT.

Since there are many entries, we used a custom Python script to extract road names and daily vehicle counts into CSV format. We have expanded the abbreviations (e.g., mapping Rd to Road, St to Street, N/S/E/W/C → North/South/East/West/Central) then converted the names to uppercase to match the names in the shapefile. The produced CSV was further manually cleaned to remove the name residuals. We have removed the road entries that have less than 10000 vehicles/day to keep only the most significant roads. The final CSV is then joined using the same method for the average speed.

The Python script and the final CSV is linked in the Appendix.

RoadName	AADT
TOLO HIGHWAY	157910
CHEUNG TSING HIGH	154800
KWUN TONG ROAD	149710
ISLAND EASTERN CO	148100
PRINCE EDWARD RO	137580
TUEN MUN ROAD	137560
TSUEN WAN ROAD	135730
GLOUCESTER ROAD	133720
VICTORIA PARK ROA	133720
HARCOURT ROAD	125100
YUEN LONG HIGHWA	122600
TSING KWAI HIGHW	120310
HONG CHONG ROAD	119980
TAI PO ROAD - SHA T	117480
FANLING HIGHWAY	114960
CHATHAM ROAD NC	114520
CHATHAM ROAD SO	113760

Figure 4: final CSV (`hong_kong_roads_aadt_cleaned.csv`) contains two columns, the road name, and the daily traffic vehicles per day.

4. Junction Blacksite Data

Junction blacksites were retrieved from the CSDI Portal, directly available as a .gdb file containing all the point locations and required no further data cleaning.

Cartographic Design Decisions

1. Map Type

The map type selected is a hybrid thematic flow line map, which is primary map type. Overlaid uniform point symbols also selected, which is secondary map type. Flow line maps are feasible for describing quantitative linear movement, such as traffic volume and along networks. To make it easy to read, we use road centrelines without arrows and imply two-way roads to avoid visual clutter. This network-style representation focuses on both lines and nodes, revealing flow distribution across Hong Kong's road system. Dot mapping for blacksites highlights distinct locations, overlap on top of the roads.

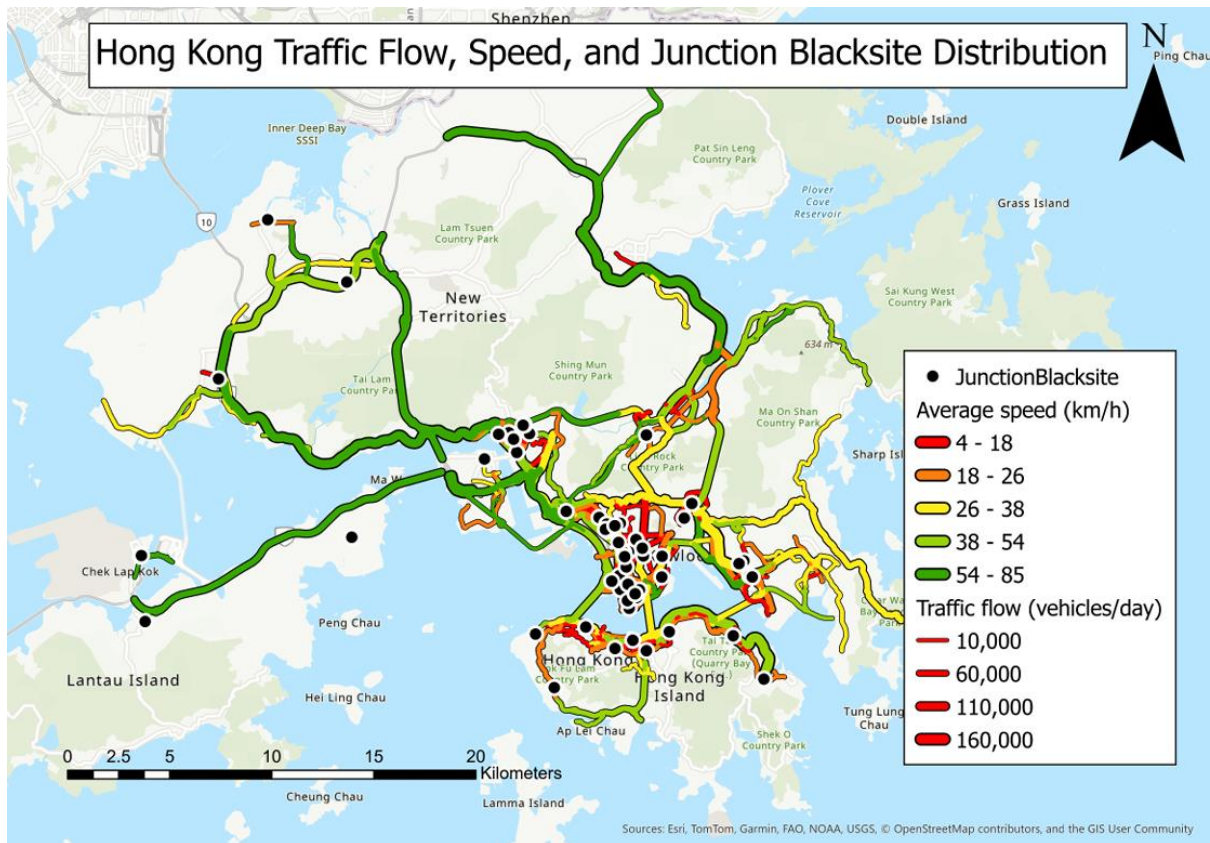


Figure 5: Our finalized map, showing the Hong Kong road networks, with temperature-like hue representing the average speed, line thickness representing the traffic flow, and points representing the junction blacksite.

2. Symbolology

Daily Traffic Flow

The line width is proportional to the daily traffic flow, classified into four equal-interval bins: 3 px (10000-60000 vehicles/day), 5 px (60000-110000), 7 px (110000-160000), and 9 px (>160000). Equal-interval classification was chosen for its simplicity and intuitive for classification (difference between classes is 50000 vehicles per day).

Average Road Speed

Line hue depicts speed, classified into five natural-breaks (Jenks) classes: 3–16 km/h (red), 16–25 (orange), 25–37 (yellow), 37–54 (lime), and 54–84 (green). Since the selected colors might blend with the base map, we have also added borders around the lines to increase the contrast. Furthermore, all the roads are dissolved into one layer, so they look interconnected regardless of the hue values. Natural breaks optimized grouping of similar values while maximizing

differences between classes, thus highlights the contrasts of road speeds across different roads. After classification, we have rounded the numbers in the labels to remove the decimal points that may cause visual clutter.

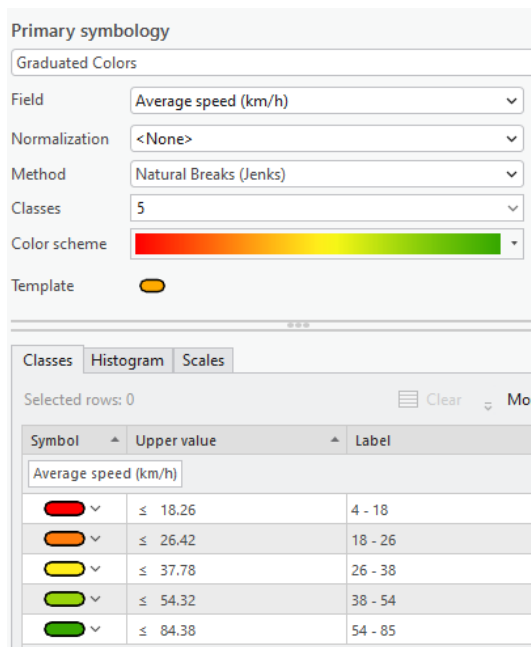


Figure 6: using the natural breaks classification method, with custom temperature color scheme and rounded number labels.

Single symbol point

Junction blacksites use uniform single-symbol point symbology: 9 pt dark gray circles with 2 pt white borders (no classification needed, as the points are nominal data). All widths exceed perception thresholds (0.1 mm minimum line thickness, 0.2 mm points as per Lecture 3), ensuring visibility and differentiation.

Legend

The sizes of the lines for traffic flow and points for the junction blacksite locations in the legend reflect the true size in the map. The map readers can match the line sizes to interpret the corresponding traffic flow easily. The line thickness for the hue was intentionally thickened to 9pt, since the hue is independent from the size, makes the colored areas larger and easier for the map readers to match the colors.

3. Primary Visual Variables - Summary

Hue and size were selected as primary visual variables for quantitative data: size for ordered magnitude (traffic flow), hue for intuitive temperature-like interpretation of speed (green = safe/fast, red = congested/slow). These align with best practices—size and value for ratio/interval data, hue universally effective. Secondary variables, such as pattern were avoided to prevent visual disturbance. The color ramp was tested for sufficient contrast (see accompanying color demonstration image showing distinct red-to-green progression against white background). Point symbols are kept in simple geometry (circles with border) for clarity, and since they were uniform data, their appearance was kept uniform and no visual variables were applied.



Figure 7: The red-green color scheme for traffic speed aligns with our daily life expectation and map viewing habits as on traffic update news. (Image source: TVB)



Figure 8: The 5 colors chosen to represent road speeds show distinctiveness and sufficient contrast.

4. Essential Map elements

Essential map elements include a map title, comprehensive legend explaining classifications and symbols, scale bar (in km for local context), north arrow and credits citing all data sources. Layout follows the planar organization principles as shown in Figure 9: optical center placement for Hong Kong, planar balance (image weighted toward center). The internal organization through strong figure-ground comparison with dark lines or points against light base and hierarchical visual planes. The result balances simplicity, functionality, and pleasant appearance, avoiding illusions or over-complexity.

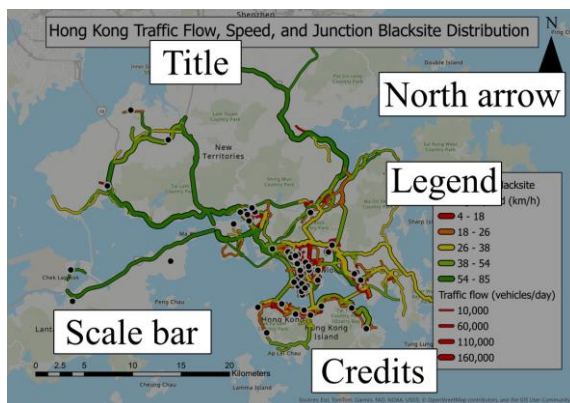


Figure 9: Our map contains the essential elements of a map.

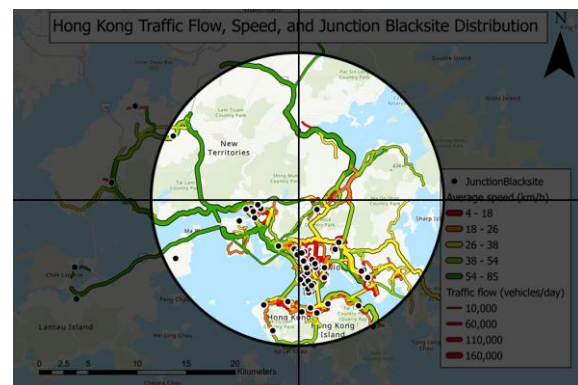


Figure 10: The Hong Kong map was shifted towards the north west to ensure the center of focus aligns with the center of the map.

Reflection on Communication Effectiveness

1. Strengths

We can evaluate our map as per content completeness, effectiveness, complexity, layout.

For content completeness, we believe the results were satisfactory. Our map contains the essential map elements. It also covers most of the Hong Kong's road networks and the major roads, analyzing the roads with more than 10000 vehicles per day, with only a few exceptions (e.g. San Tan Highway) due to the lack of source data. The sources were from the governmental traffic census 2024, thus the data is recent and reliable.

We believe that our map can effectively demonstrate traffic patterns by overlapping the junction blacksite locations on the road network maps with variable traffic flow and speeds, highlighting congestion occurs most frequently in the high volume, low speed areas, and taking safety interventions and measures should be prioritized in these areas.

Last but not least, the map appears simple and balanced. It uses intuitive symbology (width + hue + points), clear and easy-to-understand legend, and centered focus that guides viewer attention without visual overload. Contrast between line thickness, color gradients, and bordered points ensures perceptual differentiation above thresholds, while map generalization simplifies the dense road network for overview.

2. Limitations

We believe that there might be some limitations, such as the point for junction blacksite potentially clutter in ultra-dense urban areas, and thus covering the roads. This was especially prominent in Kowloon, where we have adjusted the point sizes multiple times to make the compromise between roads and junction blacksites.



Figure 11: We have also tried to use mimetic symbology (warning signs) for the junction blacksites, however the sizes were set to 17 pt to be able to view the inner triangles from a distance, and it created visual clutter. The finalized map used only 9pt (7pt inner dot and 2pt border), significantly smaller and avoids blocking the roads in Kowloon.

Overall, it is believed that the map still achieves its goal of supporting data driven decisions, and its strengths outweigh its limitations, proving the real-world applications of thematic maps in cartography.

Task Allocation Among Teams

Student Name	Task
██████████ ██████████	Theme selection, report writing, ArcGIS Pro processing, symbology design and classification, PowerPoint making
██████████ ████████████████	PowerPoint making, report writing
██████████ ██████████	PowerPoint making
██████████ ██████████	PowerPoint making
██████████ ██████████	Theme selection, report writing, ArcGIS Pro processing, symbology design and classification, PowerPoint making
██████████ ██████████	Theme selection, data collection and cleaning, ArcGIS Pro processing, symbology design and classification, report writing, PowerPoint making

References

Tomtom (2025). Hong Kong traffic in 2025. Retrieved from

<https://www.tomtom.com/traffic-index/city/hong-kong>

Transport Department (2021). *Data Specification for Junction Blacksite List*. Retrieved from

https://www.td.gov.hk/filemanager/en/content_2042/.opendata/junctionblacksite_dataspec_en.pdf

Transport Department (2024). *Annual Traffic Census 2024*. Retrieved from

<https://atc.td.gov.hk/2024/AnnualTrafficCensus2024.pdf>

Appendix A – Data Sources for Map

Data.gov.hk. *Annual traffic census traffic data (AADT)*. Retrieved from

https://data.gov.hk/en-data/dataset/hk-td-tis_7-traffic-flow-census/resource/c3392d0f-9a7c-41ad-9b0c-cfe8c8334f51

Data.gov.hk. *Road centerline (shape and names of roads)*. Retrieved from

https://data.gov.hk/en-data/dataset/hk-td-tis_7-traffic-flow-census/resource/a2009821-77c7-4788-815f-932d595bc4fb

Data.gov.hk. *Hong Kong car journey time data (travel speed)*. Retrieved from

https://data.gov.hk/en-data/dataset/hk-td-tis_34-car-journey-time-data

CSDI Portal. *Junction blacksite list*. Retrieved from

https://portal.csdi.gov.hk/geoportal/?lang=en&datasetId=td_rcd_1638947917303_26860

Appendix B – Finalized Average Speed Data (Whole Version in excel file)

[illegible]

Appendix C – Finalized Annual Average Daily Traffic Data (Whole Version in excel file)

RoadName	AADT
2 TOLO HIGHWAY	157910
3 CHEUNG TSING HIGHWAY	154800
4 KWUN TONG ROAD	149710
5 ISLAND EASTERN CORRIDOR	148100
6 PRINCE EDWARD ROAD EAST	137580
7 TUEN MUN ROAD	137560
8 TSUEN WAN ROAD	135730
9 GLOUCESTER ROAD	133720
10 VICTORIA PARK ROAD	133720
11 HARCOURT ROAD	125100
12 YUEN LONG HIGHWAY	122600
13 TSING KWAI HIGHWAY	120310
14 HONG CHONG ROAD	119980
15 TAI PO ROAD - SHA TIN	117480
16 FANLING HIGHWAY	114960
17 CHATHAM ROAD NORTH	114520
18 CHATHAM ROAD SOUTH	113760
19 LUNG CHEUNG ROAD	113670
20 KWAI CHUNG ROAD	109870
21 TSEUNG KWAN O ROAD	109680
22 CHING CHEUNG ROAD	108590
23 KWUN TONG BYPASS	102550
24 CANAL ROAD	101690
25 CANAL ROAD FLYOVER	101690
26 WAN CHAI INTERCHANGE	101690
27 WEST KOWLOON HIGHWAY	100050
28 LAI CHI KOK BAY BRIDGE	99870
29 PRINCESS MARGARET ROAD	96840
30 WESTERN HARBOUR CROSSING	96830
31 CONNAUGHT ROAD CENTRAL	95630
32 CROSS HARBOUR TUNNEL	94820
33 LAI CHI KOK ROAD	94700
34 TSING LONG HIGHWAY	94700
35 GASCOIGNE ROAD	94070
36 GASCOIGNE ROAD FLYOVER	94070
37 LANTAU LINK	91370
38 WATERLOO ROAD	91130
39 LEI YUE MUN ROAD	89960
40 SAN TIN HIGHWAY	89230
41 LION ROCK TUNNEL	86980
42 LION ROCK TUNNEL ROAD	85450
43 TSEUNG KWAN O TUNNEL	84800
44 TSEUNG KWAN O TUNNEL ROAD	84800
45 TATE'S CAIRN HIGHWAY	83770
46 TATE'S CAIRN TUNNEL	83770
47 MA ON SHAN ROAD	82740
48 NORTH LANTAU HIGHWAY	82160
49 CHEUNG TSING BRIDGE	79660
50 CHEUNG TSING TUNNEL	79660
51 CHEUNG TSING TUNNEL ROAD	79660
52 CLEAR WATER BAY ROAD	78470
53 PRINCE EDWARD ROAD WEST	76800
54 FERRY STREET	75180
55 TEXACO ROAD	75030
56 SHA TIN ROAD	74840
57 MA TAU CHUNG ROAD	73160
58 SHING MUN TUNNEL ROAD	72220
59 EAST KOWLOON CORRIDOR	70960
60 KAI LUK ROAD	69740
61 LIN CHEUNG ROAD	69040
62 WONG CHU ROAD	68950
63 EASTERN HARBOUR CROSSING	68800
64 WEST KOWLOON CORRIDOR	66710
65 MORRISON HILL ROAD	64830
66 STONECUTTERS BRIDGE	64520
67 NAM WAN TUNNEL	64290
68 EAGLE'S NEST TUNNEL	63850
69 WONG CHUK HANG ROAD	63610
70 KAI TAK TUNNEL	63500
71 QUEENSWAY	62970

Appendix D – Link to miscellaneous files (.csv, .xml, .xlsx, .py files for data cleaning)

https://connectpolyu-my.sharepoint.com/:f/g/personal/24001355d_connect_polyu_hk/IgAueNQ3-fsuSYwDx90KtHhNAb7W7OB9c_ND5tz6cYnvepk?e=oJvMie

File name	Description
ATC_TRAFFIC_DATA.zip	Original zip archive for traffic flow data, downloaded from data.gov.hk
ATC_TRAFFIC_DATA_2024_GISDB.xml	Renamed from GISDB.xml, the xml file containing the traffic flow data of 2024.
AverageRoadSpeed_2024(AverageSpeed).csv	Exported CSV from its excel spreadsheet.
AverageRoadSpeed_2024.xlsx	Contains raw spreadsheet data of traffic speed from data.gov.hk, and calculations.
hong_kong_roads_aadt_cleaned.csv	Manually cleaned CSV after aggregating with Python script.
hong_kong_roads_aadt.csv	CSV file directly after Python script conversion from XML to CSV.
JunctionBlacksiteList_FGDB.zip	Original zip archive for junction blacksites, downloaded from CSDI portal.
main.py	Python script that converts XML to CSV.
RoadCentreline_FGDB.zip	Original zip archive for road networks, downloaded from CSDI portal.